

2.10.54

EE25BTECH11025 - Ganachari Vishwambhar

Question:

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be unit vectors such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$. Which of the following are correct?

- 1) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} = \mathbf{0}$
- 2) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq \mathbf{0}$
- 3) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{c} \neq \mathbf{0}$
- 4) $\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}$ are mutually perpendicular.

Solution:

Given:

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0} \quad (1)$$

$$\mathbf{c} = (\mathbf{a} \quad \mathbf{b}) \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad (2)$$

$$((\mathbf{a} \quad \mathbf{b})) = A \quad (3)$$

Proving that A is non-singular (assuming 2D space):

For the matrix to be singular, the vectors in the matrix should be linearly dependent, i.e., collinear. The determinant of the matrix should be zero.

Multiplying (1) $\mathbf{a}^\top, \mathbf{b}^\top, \mathbf{c}^\top$. We get

$$\mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{c} = \mathbf{c}^\top \mathbf{a} = \frac{-1}{2} \quad (4)$$

$$(5)$$

The angle between the vectors taken pairwise is 120° . Hence, the cross product between the vectors is given by:

$$|A| = |(\mathbf{a} \quad \mathbf{b})| = |\mathbf{a}||\mathbf{b}| \sin \theta = 1.1. \sin 120^\circ = \frac{1}{2} \neq 0 \quad (6)$$

Hence, the given matrix A is non-singular.

Similarly, we can show the same for vectors $\mathbf{a}$ and $\mathbf{b}$.

Hence, the correct option is (3) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq \mathbf{0}$

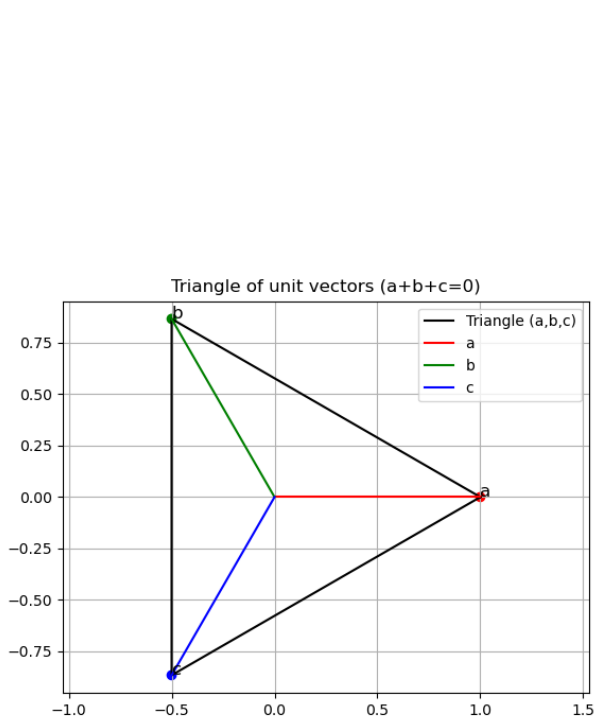


Fig. 1: Plot of the $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$